

Physics-First Formalization of the Strüver Model

Ontology-First Axioms, Domain-Invariant Structure, and Falsification Protocols
(Public Layer: Ontological Definitions and Observable Projections)

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Contents

1	Purpose and Methodological Standard	2
1.1	Goal	2
1.2	Core separation principle	2
2	Ontological Primitives (Domain-Invariant)	2
2.1	Axiom E (Energy as transition-capacity)	2
2.2	Axiom M (Mass as energy-density)	3
2.3	Axiom R (Space as relation between energy-densities)	3
2.4	Axiom P (Process / ordering; “time” as operator)	3
3	State, Constraint, and Stability (Abstract Physics Form)	3
3.1	Physical state	3
3.2	Instability and calibration	3
3.3	Dynamics with corridor stability	3
3.4	Lyapunov-style requirement (stability criterion)	4
4	TRIAS Structure as a Domain-Invariant Decomposition	4
4.1	Corridor (70/30 as a dynamic corridor, not a fixed point)	4
5	Admissibility, Refusal, and External Falsifiability (Public Semantics)	4
5.1	Admissibility as a closed-world constraint	4
5.2	Epistemic refusal	4
5.3	Decision semantics (public)	5
5.4	Frozen protocol requirement	5
6	Observable Projections (Physics Tests Without “Spacetime” Postulate)	5
6.1	Propagation-through-medium principle (public)	5
6.2	Ringdown ordering discriminator (minimal invariant projection)	5
6.3	Notes on domain freedom vs invariant structure	6
7	Mapping Layer (Public Interface, Domain-Specific Content)	6
7.1	General mapping template	6
7.2	Domain examples (non-exhaustive)	6
8	External Verification: What a Third Party Can Validate	6
9	Appendix A: Minimal Audit Record (Public Field Semantics)	7

License and Disclosure Boundary

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Scope of this document: This text provides (i) ontological definitions, (ii) domain-invariant constraints, and (iii) publicly testable *observable projections* and falsification criteria.

What is intentionally not disclosed: the full operator architecture, numerical extraction pipeline, implementation details, and the complete admissibility kernel. Those are reviewed only under NDA in a controlled technical setting.

Non-reconstructability claim: The relations in this document are designed to be *non-invertible* with respect to proprietary implementation.

1 Purpose and Methodological Standard

1.1 Goal

This document “physicizes” the Strüver model by binding its primitives to observable constraint, while keeping implementation details protected. The standard is:

- **Ontology first:** define what exists and how it relates.
- **Operational binding:** connect primitives to measurable constraints.
- **Falsifiability:** provide rejection conditions under frozen protocols.
- **Domain invariance:** the structural core must not drift across domains.

1.2 Core separation principle

We distinguish:

1. **Invariant layer** (public): axioms, definitions, and observable projections.
2. **Mapping layer** (domain-specific): proxy selection, calibration, embedding into instruments.
3. **Kernel layer** (proprietary): deterministic admissibility operators and commit-time gating.

Only (1) is fully public here.

2 Ontological Primitives (Domain-Invariant)

We define three primitives as *ontological commitments*. They are not merely units or formulas; they are claims about what a physical theory is about.

2.1 Axiom E (Energy as transition-capacity)

Axiom E: Energy is the capacity for state transition under constraints.

$\mathcal{E} \equiv$ “transition-capacity” $\Rightarrow$ $\mathcal{E}$ is operationally expressed by the ability to produce change (work) under (1)

Operational binding: For any closed experimental protocol, energy is inferred through conserved exchange channels (mechanical, electromagnetic, thermal, etc.). Units (J) are bookkeeping, not ontology.

2.2 Axiom M (Mass as energy-density)

Axiom M: Mass is energy-density (bound transition-capacity per relational extent).

$$\mathcal{M} \equiv \rho_{\mathcal{E}} \text{ (energy-density)}. \quad (2)$$

Operational binding: Inertia (resistance to acceleration) and binding profiles are empirical expressions of localized energy-density. This is compatible with mass-energy equivalence as an identity between two bookkeeping forms of the same primitive.

2.3 Axiom R (Space as relation between energy-densities)

Axiom R: Space is not a container; it is a relational structure among energy-density distributions.

$$\mathcal{R} \equiv \text{“relational structure between energy-densities”}. \quad (3)$$

Operational binding: Distances are derived from propagation/interaction relations (signal delay, coupling strength, exchange constraints). No ontological postulate of “spacetime” is required.

2.4 Axiom P (Process / ordering; “time” as operator)

Axiom P: “Time” is treated as an ordering/progression operator associated with transition and stabilization, not as an ontological dimension.

$$\Phi : \text{orders transitions and corridor return, i.e., it is process, not background geometry.} \quad (4)$$

This does *not* deny clock readings; it states clock readings are measurement proxies for process regularities.

3 State, Constraint, and Stability (Abstract Physics Form)

3.1 Physical state

Let the physical state be an element of a constrained state space:

$$x \in \mathcal{X}, \quad \mathcal{X} \text{ defined by conservation and admissibility constraints.} \quad (5)$$

3.2 Instability and calibration

We define instability as a nonnegative functional:

$$B(x) \geq 0, \quad (6)$$

and calibration as a control-like operator that pushes the state back toward a stable corridor:

$$u(x) = \mathcal{K}(x, B). \quad (7)$$

3.3 Dynamics with corridor stability

We express dynamics generically as:

$$\dot{x} = f(x) + g(x) u(x), \quad (8)$$

and define a corridor $\mathcal{C} \subset \mathcal{X}$ such that:

$$x \in \mathcal{C} \Rightarrow \text{admissible stable operation}, \quad x \notin \mathcal{C} \Rightarrow \text{drift or dominance regimes.} \quad (9)$$

3.4 Lyapunov-style requirement (stability criterion)

There exists a scalar functional $V(x) \geq 0$ such that:

$$\dot{V}(x) \leq 0 \quad \text{for admissible correction trajectories.} \quad (10)$$

This is the minimal mathematical expression of “corridor return” without revealing implementation.

4 TRIAS Structure as a Domain-Invariant Decomposition

The Strüver model uses a triadic decomposition:

$$u = (u_X, u_Y, u_Z), \quad u_i \geq 0, \quad (11)$$

with normalized shares

$$S = \sum_i u_i, \quad \rho_i = \frac{u_i}{S}, \quad \sum_i \rho_i = 1. \quad (12)$$

Interpretation in physics:

- X : structural capacity (constraint enforcement / load-bearing)
- Y : coherence capacity (coupling integrity / non-fracture)
- Z : adaptive capacity (resolution / measurement / correction bandwidth)

These labels are *role-based* and domain-invariant.

4.1 Corridor (70/30 as a dynamic corridor, not a fixed point)

We treat the “70/30” operator as a *calibration corridor* rather than a fixed value:

$$p^* \in (0, 1), \quad p \in [p_{\min}, p_{\max}] \approx [0.20, 0.40], \quad (13)$$

and more generally as a dynamic admissible band (typical corridor interpretation: $\sim 60\text{--}80\%$ freedom vs. $40\text{--}20\%$ order), domain-calibrated but structurally fixed.

5 Admissibility, Refusal, and External Falsifiability (Public Semantics)

5.1 Admissibility as a closed-world constraint

Define an admissible set $\mathcal{A} \subseteq \mathcal{X}$:

$$x \in \mathcal{A} \iff \text{all required premises are measured/resolved and constraints are satisfied.} \quad (14)$$

5.2 Epistemic refusal

If required premises are absent or measurement-underdetermined, execution must be refused:

$$\text{If } \exists \text{ required premise } \pi \text{ such that } \pi(x) \text{ is unresolved} \Rightarrow \text{REFUSE.} \quad (15)$$

Public meaning: refusal is a safety primitive at the irreversible boundary; it is not “uncertainty” as a probability score, but a discrete admissibility veto.

5.3 Decision semantics (public)

At a high level:

$$\text{PASS} \Rightarrow \text{ACCEPT}, \quad \text{WARN} \Rightarrow \text{ADJUST}, \quad \text{FAIL} \Rightarrow \text{REJECT (or REFUSE if epistemically underdetermined)} \quad (16)$$

This describes behavior without revealing internal gates.

5.4 Frozen protocol requirement

Any claim is only meaningful under:

- pre-registered extraction protocol,
- frozen parameterization (no per-event tuning),
- public rejection corridor and uncertainty accounting.

6 Observable Projections (Physics Tests Without “Spacetime” Postulate)

This section states *testable* projections that do not require publishing the proprietary kernel.

6.1 Propagation-through-medium principle (public)

If gravitational effects are propagation of energy-relation change through an energy-structured relational medium (Axiom R), then propagation can depend on background energy-density along the path. This yields a falsifiable class:

$$\text{Propagation}(d, z) \neq \text{pure geometric scaling only.} \quad (17)$$

6.2 Ringdown ordering discriminator (minimal invariant projection)

Define the ringdown decay time τ and characteristic frequency f . A minimal ordering relation is:

$$\tau \propto \frac{1}{f}, \quad \tau = \frac{1}{2\pi C_S f}, \quad (18)$$

where C_S is fixed *a priori* (not tuned per event).

What this is (and is not).

- **Is:** a domain-level observable projection of an invariant ordering timescale associated with corridor return.
- **Is not:** the full dynamical envelope of each event, nor a claim that all events are dynamically identical.

Rejection condition. Under a frozen pipeline and pre-registered protocol, if Eq. (18) shows no systematic alignment within a defined uncertainty corridor across events, the framework is rejected for that domain.

6.3 Notes on domain freedom vs invariant structure

The model allows:

- domain-specific dynamical variance (source parameters, modes, couplings),
- but enforces a lower-order invariant structure (ordering relation / admissibility conditions).

7 Mapping Layer (Public Interface, Domain-Specific Content)

The mapping layer binds roles (X, Y, Z) and state variables to measurable proxies.

7.1 General mapping template

Invariant role	Domain proxy (example)	Measurement channel
X (structure)	load-bearing constraint / binding	stress/strain, stability margins
Y (coherence)	coupling integrity / phase coherence	coherence metrics, residual drift
Z (adaptation)	resolution / sensing bandwidth	SNR, sampling, calibration capacity

Important: mapping choices are falsifiable and may be revised; the invariant role structure is not.

7.2 Domain examples (non-exhaustive)

- **Robotics:** X joint torque envelope, Y control coherence, Z sensing/correction resolution.
- **Satellites:** X structural/thermal/power margins, Y cross-subsystem coherence, Z measurement/fault-resolution bandwidth.
- **Seismics:** X energy release constraints, Y wavefield coherence, Z measurement resolution and inversion capacity.

8 External Verification: What a Third Party Can Validate

A third party can validate *without access to the proprietary kernel*:

- refusal semantics (premise missing $\Rightarrow$ refusal),
- append-only audit semantics (canonical record, hashing),
- stability corridor conformance on observable projections,
- reproducibility under frozen extraction pipelines.

They cannot reconstruct:

- internal operator definitions,
- gate thresholds and their construction,
- the kernel's complete admissibility topology.

9 Appendix A: Minimal Audit Record (Public Field Semantics)

A minimal audit record (illustrative, not implementation):

- context identifier (domain, scenario),
- inputs (feature representation; redacted where needed),
- decision outcome (ACCEPT/ADJUST/REJECT/REFUSE),
- refusal rationale category (e.g., “missing premise”, “underdetermined measurement”),
- hash reference for integrity.

10 Appendix B: A compact “physics-first” summary

Energy is transition-capacity. Mass is energy-density. Space is the relational structure among energy-densities. “Time” is the ordering of transition and stabilization. The invariant triadic structure enforces corridor stability and refuses execution under underdetermination. Observable projections provide falsifiable discriminators under frozen, pre-registered protocols.